Real-time-based Object Recognition using SIFT algorithm

Dhandapani Vaithiyanathan[a*[0000-0001-5235-2620]], **Muniraj Manigandan**[b]
[a]*Department of Electronics and Communication Engineering*, National Institute of Technology, Delhi, India
[b]*School of Electronics Engineering*, Vellore Institute of Technology, Chennai, India

[a*]dvaithiyanathan@nitdelhi.ac.in, [b]manigandan.m@vit.ac.in

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***Abstract*— In this paper, we propose a real-time object recognition system that makes use of local image features using Scale Invariant Feature Transform (SIFT). The features are partial changes in illumination and affine, and they are invariant to rotation, image scaling, and translation. Similar characteristics are shared by these features and the inferior temporal cortex neurons involved in object recognition. Features can be efficiently detected using a phased filtering technique in scale space. Image keys generally support the local geometric information representing the blurred image gradient at different scales and orientation planes. An indexing technique called nearest-neighbor search (NSS) uses the keys as input to determine the object matches. Based on the number of descriptors we categorize the object is recognized or not, and the final output is displayed in the microcontroller-based display unit.**

Keywords*— *Object Recognition, SIFT, Nearest Neighbor Search, Difference of Gaussian (DoG)

I. INTRODUCTION

Local image properties undisturbed by neighboring clutter or partial occlusion are necessary for object recognition in congested real-world environments. The characteristics must be at least somewhat unaffected by 3D projective transforms and lighting on the object which modifies the image details. The image features, also known as key point features or interest points, are intriguing regions (such as edges, corners, and blobs) that are relatively specific to that particular image. We can analyze, describe, and match images in the context of image alignment, robot navigation, object recognition, motion tracking, and other applications by analyzing the image, identifying a set of key points, and computing a descriptor vector or feature vector for each key point. The following is a list of some of the most used feature detection methods:

SURF (Speeded Up Robust Features) is a computationally expensive and quite effective algorithm. SURF is the extension of SIFT algorithm. Moreover, the SURF algorithm dealt with orientation using the sum of the Haar wavelet responses and uses the Hessian matrix approximation to find interesting points, especially for object detection and recognition [1]. Only key point detection is done in real-time applications using FAST (Features from Accelerated Segment Test). There is no descriptor, and we can compute the descriptors using SIFT or SURF [2]. A faster way of matching and computing feature descriptors is BRIEF (Binary Robust Independent Elementary Features). BRIEF does not provide a mechanism to locate the features; it merely acts as a feature descriptor [3].

FAST and BRIEF are combined in ORB (Oriented FAST and Rotated BRIEF) [4]. The feature descriptor HOGs (Histogram of Oriented Gradients) has been widely and successfully used for object detection. It displays things as a SINGLE feature vector rather than a collection of feature vectors, where each one represents a section of the image. HOGs are calculated using a sliding window detector and merged into a single feature vector to represent the image. For each place in the window, a HOG descriptor is computed [5]. Scale invariance is a problem in computer vision, and SIFT (Scale Invariant Feature Transform) efficiently solves it. Interesting important points are discovered using the Gaussian Difference technique [6].

This paper presents a real-time object recognition for image feature generation using SIFT approach. With this method, an image is split up into a number of small local feature vectors, each of which has some degree of resistance to variations in light, affine or 3D projection, as well as to the translation, scaling, and rotation of the image. Additionally, this study discusses the enhanced model verification and indexing methodologies. A filtering method effectively identifies the scale-invariant features. The first step is to locate significant locations in grid points by searching the NSS that are maxima or minima of a DoG function. Every single feature describes the local image region sampled especially in the scale-space coordinate. These keypoint features accomplish partial consistency to local alterations, such as affine or 3D projections, where the detail of specific NSS are hidden in the image gradients. This approach is predicated on a simulation of the complex cell activity in the human visual cortex. SIFT keys are the generated feature vectors. After the model has been created, we utilize it to recognize objects, and we can also reasonably estimate their pose.

The paper is organized as follows: the related works are briefed in Section II, and Section III provides a brief description of the methodology for the real-time object recognition system using SIFT approach that is suggested. Section IV discusses the discussions and outcomes of the experiments, and Section V provides a summary of the results obtained.

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II. RELATED WORKS

A salient point detector based on scale space theory and the difference of Gaussian in [7]. In [8], a Fast-Hessian interest point detector, that sampling on a regular dense grid outperforms complex salient point detectors producing much more image patches than interest point operators. However, a method is suggested in [9, 10] that makes it necessary for the singular target object to take up the majority of each of the training view images, making it practically meaningless. The other most crucial methods involve identifying and describing the stable surface characteristics of the object from 2D view images, such as contour features, texture, shape, and color. The SURF algorithm is used for feature extraction and object recognition in [11], which also provides additional details on the algorithm. Their research demonstrates that the query image is used as an input image before the object to be recognized is extracted Additionally, [12] describes in detail an advanced method of image identical matching using SIFT, which is a way to distinguish an image from the images that have already been saved in the database. This paper presents a well-known image-matching method SIFT for real-time object recognition. Some recent advancements and the applications based on SIFT are mentioned in [13 -16]

Two frequently used image patch descriptors are the SURF descriptor [1] and SIFT descriptor [7]. They are both independent of light and rotation. The object identification issue is addressed using the SIFT [7] key points descriptors. The capacity to identify particular instances or general classes of objects is a topic of growing interest in object recognition research. We concentrate on the issue of identifying particular examples of objects in the images. Our approach recognizes the object's using an object recognition approach with the visual object model details. The method takes SIFT key points and descriptions as input to find possible object matches using a nearest-neighbor indexing method. To avoid this issue, SIFT is used to acquire the object model for numerous views and images of the same object. Our method recognizes objects by matching their key points by only matching the key points of the query image with the test image having similar key points as those of the object models.

III. PROPOSED METHODOLOGY

In this paper, the proposed methodology is shown in Figure 1. Initially, the single image is captured from a sequence of video frames and stored in the database. Further, the single is compared with the real-time image to recognize the objects present in the image scene, and the object recognition is carried out based on SIFT algorithm with necessary steps as shown in Figure. For object recognition, the key features are obtained based on the detection as well the number of the descriptor. If the object is recognized the microcontroller includes a decision-making rule to display the recognized object in the LCD. Further, the detailed SIFT object recognition description is briefed in this section.

A. SIFT

To match an item or scene in the image, the SIFT algorithm aims to build a database of features with a variety of properties. The keypoint features are invariant to various parameters such as image rotation, change in lighting, and scaling as well as partly invariant to changes in 3D camera viewpoint. With the help of effective algorithms, more characteristics can be extracted from all similar images. Moreover, the keypoint features are very distinct for every image, making it possible to accurately match a single feature against a vast database of features with a high probability. The algorithm's main steps are as follows:

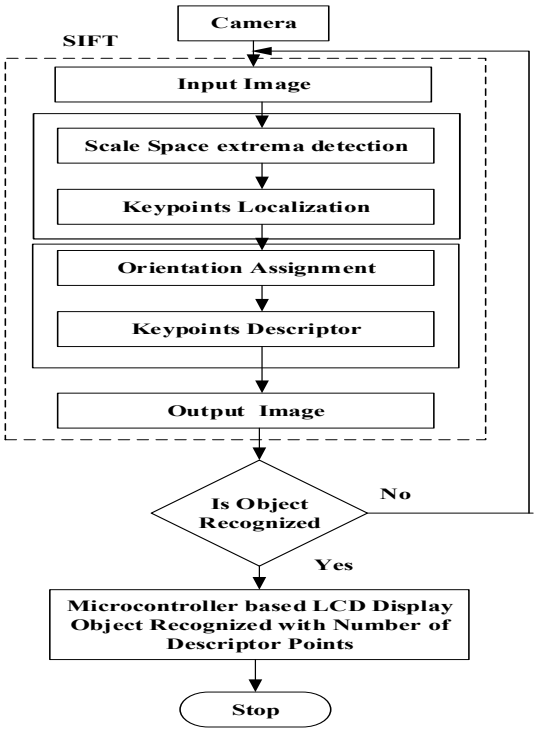


Fig. 1. Proposed Methodology

1. Scale-space extrema detection: The DoG function locates possible concerned points which are orientation-invariant.
2. Key-point localization: This localization is to establish location and scale that helps to measure the key point's stability.
3. Orientation assignment: The local image gradient directions are assigned to each keypoint position.
4. Keypoint descriptor: This descriptor is obtained from the region surrounding and the corresponding local image gradients measured that depend on the selected scale. Further, the scaled image is transformed into a symbol that permits the substantial representation of the light shifting based on image gradients with local shape distortion.

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The Algorithm suggests to build a DOG structure as can be shown in Figure 2:

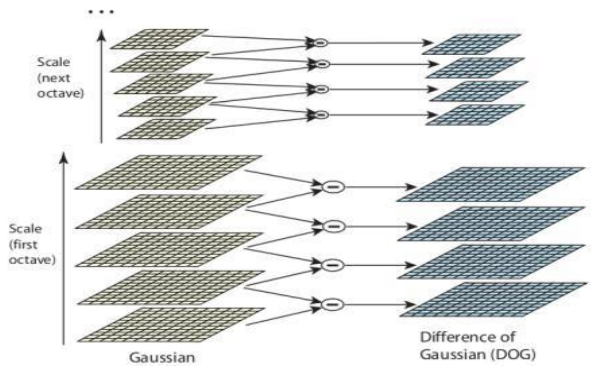


Fig. 2. The DOG pyramid

To reduce noise, the original image is first filtered with a Gaussian function. The keypoint detection, together with the results, demonstrate that localization repeatability keeps increasing as the DOG pyramid is generated by comparing the NSS. The maximum and lowest descriptor detection are shown in Figure 3 respectively. In addition to the first and final DOG at each scale, we search maxima. A key point is represented by each maximum. The coordinate of the key point is determined for each scale using the downscaling factor. Each key point's scale is recorded for further use. Thereafter, each image experienced a variety of adjustments, such as rotation, scaling, affine stretching, and alterations to the brightness and contrast may lead to image noise and loss in the image details. Figure 4 illustrates how the relationship between the number of key points and the number of scales per octave increases up to an asymptote. Yet, there are roughly 80% of critical points are repeated, making this number roughly constant.

1) Key-point localization

a) Contrast threshold

Once the key points were set, we would like to remove all key points that are not stable enough. The first step is to set a threshold of the contrast with its environment. The DoG function is a kind of contrast function. So, all key points with a value in DoG less than 0.03 (after normalizing the intensity [0 1]) are removed from the key points list.

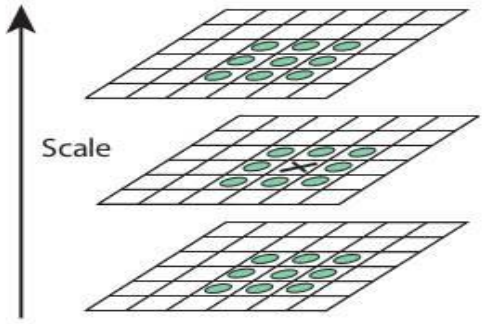


Fig. 3. Maxima and Minima Detection

b) Eliminating edge response

The second method for key-point localization is to eliminate edge responses. To do that, we use the Hessian of the DoG function,

$$H = \begin{bmatrix} D_{xx} & D_{xy} \\ D_{yx} & D_{yy} \end{bmatrix}$$

For instance, due to variations in the DoG function, there is a major change in the curvature at the edges which are in the perpendicular direction, and we want to exclude all key points that lack a stable maximum. Therefore, using the prediction of H and their product, we can get the sum of the Eigenvalues.

$$T_r(H) = D_{xx} + D_{yy} = \alpha + \beta$$

$$Det(H) = D_{xx}D_{yy} - D_{yx}D_{xy} = \alpha\beta$$

The curvatures have different signs in rare circumstances whenever the determinant is negative. Let r represent the proportion of the largest to the smallest magnitude Eigenvalue. Then,

$$\frac{T_r(H)}{Det(H)} = \frac{(\alpha+\beta)^2}{\alpha\beta} = \frac{(r\beta+\beta)^2}{r\beta^2} = \frac{(r+1)^2}{r}$$

$$\frac{T_r(H)^2}{Det(H)} < \frac{(r+1)^2}{r}$$

Keypoints with a ratio between the primary curvatures greater than 10 are disregarded when r = 10 as mentioned in [1].

2) Orientation assignment

The keypoint descriptor is nothing but the relative orientation which leads to becoming rotation-invariant by giving each keypoint a fixed orientation depending on local image attributes. A circular window with a Gaussian weight 1.5 times the key-point scale to enhance each gradient. Peaks in the orientation histogram overwhelm by the local gradients and their directions. After the highest peak in the histogram has been identified, the highest peak is used to build a key point with that orientation.

3) Keypoint descriptor

We will recover all the information obtained on each point (orientation, scale, and location) in a descriptor after identifying all potentially invariant points. The gradient magnitude and orientation at each image sample point are first calculated to construct a key-point description. The circle put on top of them indicates that they are weighted by a Gaussian window. For experimental analyzes use 4x4 descriptors are computed from a 16x16 set of samples. Figure 4 depicts a 2x2 descriptor array using an 8x8 set of samples. To generate the descriptor, a vector is constructed from the values of each entry corresponding to the lengths of the arrows and each entry is the orientation histogram highlighted by the arrows, as shown in Figure 4.

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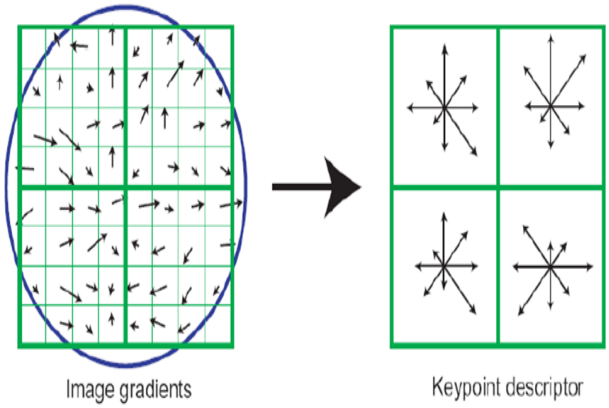


Fig. 4. Construction of descriptors

B. *Matching algorithm*

We must identify the SIFT point for each frame to compare two frames and find a match. The key point descriptors are used by the matching algorithm for matching. We look for key points in the following frame that correspond to each key point in the preceding frame. The best potential match for each key point is discovered by locating each key point's nearest neighbor in the key points. The value of this metric for actual image data is depicted in Figure 5. The ratio of each key point's closest to second-closest neighbors is used to represent the probability density curves for correct and erroneous matches. We used a matching technique to compare two images with a ratio of 0.6. In this instance, we marked the book in the left image and used the right image to run the matching algorithm. The key point in the right image is connected to matches from the left image via the green line. The SIFT key points are found at edge points, as is evident. Every key point has a blue arrow that indicates the direction in which it is oriented concerning that of the feature key points.

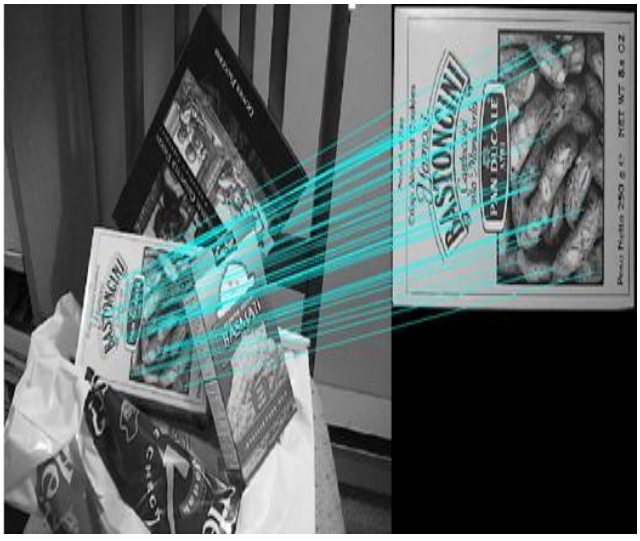


Fig. 5. Image matching based on SIFT

a) *Descriptor representation*

Before deciding on the level of Gaussian blur for the image, the magnitudes and orientations of the image gradients are sampled around a key point. The gradients are already computed for each level of the pyramid to perform effectively. To lessen the effects of changing illumination, the feature vector is finally normalized. By reducing the gradient magnitudes based on the thresholding values in each of the unit feature vectors that are no greater than 0.2 and then normalizing it to unit length. As a result, it is no longer as crucial to match the magnitudes for big gradients, and the distribution of orientations is now given more weight. Using various illuminations for the same items, the empirically determined value of 0.2 was obtained.

b) *Descriptor testing*

The two variables that can be changed to alter the complexity of the descriptor are the number of orientations (r) in the histograms and the breadth (n) of the n × n array of orientation histograms. The major limitation is that as the descriptor's complexity increases more susceptible to shape distortions, noise, and occlusion in the image details causing less accuracy.

C. *Hardware setup for Object Recognition*

Figure 6 shows the hardware setup for displaying the detected object based on the decision-making rule ported into the microcontroller. The decision is based on the maximum number of descriptors attained by comparing the test image and the images from the database. If the real-time test image has a maximum descriptor, then the result will be displayed as “object recognized” and the name of the object being detected is also displayed along with several descriptor points which are considered key feature points.

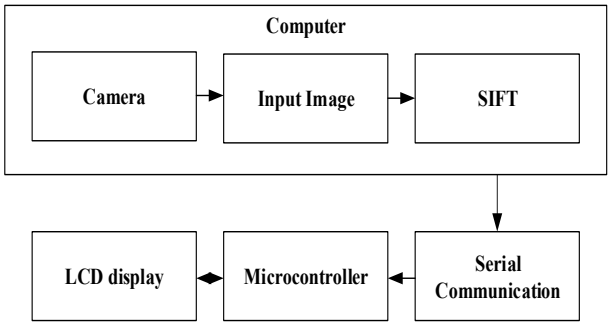


Fig. 6. Hardware configuration for displaying the detected object and number of descriptors.

IV. Experimental Results and Discussions

Figure 7 shows the output for different image orientations for two non-identical images. Figure 8 shows output verified for different sizes of the input image with the different illumination is verified with images stored in the database. Table 1 indicates the non-object recognition and object recognition with a number of descriptor key points.

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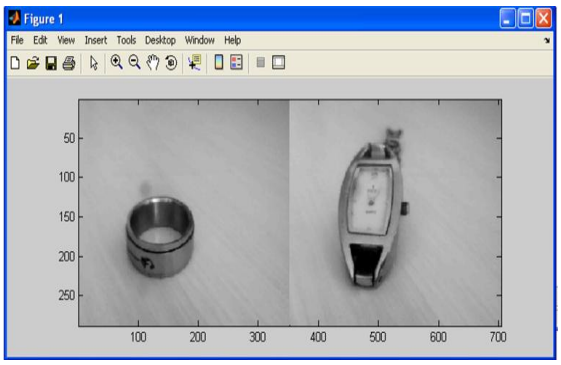


Fig. 7. Matching between Two non-identical images ring is the real-time test image and the watch is the image from the database.

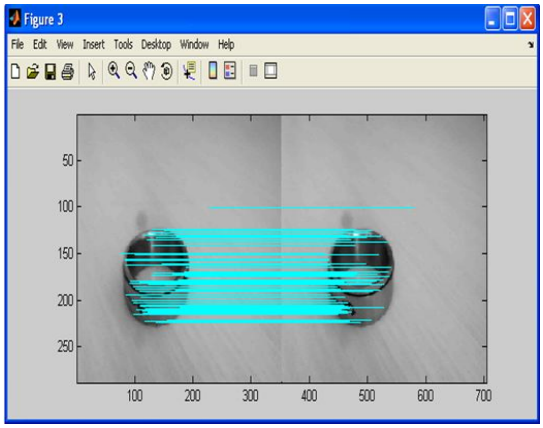


Fig. 8. Matching between Two identical images ring is the real-time test image and the watch is the image from the database. The blue lines indicate the maximum descriptor key points matching for object recognition.

TABLE I. OBJECT RECOGNITION – MATCHING DESCRIPTION

Image	SIFT Object Recognition		
	Real-time image	***Comparison of real-time with the dataset***	***Matching Description***
Image - Ring	69 key points for Ring	150 key points for watch image	No match found
Image - Ring	69 key points for Ring	69 key points for ring image	Match found

V. CONCLUSION

We have demonstrated an approach to object recognition that is based on local feature-based object modeling. We used an object multiview dataset for this challenge. The technique executes modeling while only choosing a portion of the views, resulting in a condensed representation. Views are chosen by comparing SIFT descriptors to compare how similar they are to each other. By a query image is compared to the extracted models, recognition is accomplished with fewer comparisons than with the complete dataset. With all available views of the item, only the pose estimate phase needs to be carried out. Our approach performs well in terms of object recognition and posture estimation accuracy. By utilizing datasets with numerous degrees of freedom or object detection and modeling in the video, the method utilized in this paper may be easily expanded. It would also be intriguing to apply our research in a situation with several local keypoint descriptors.

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